Praktikum Basis Data

Modul 08

Scripting_SQL_(Create_index,_Create view,_Analyze_Table)

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK
FAKULTAS INFORMATIKA
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JURNAL

1.

a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.

```
CREATE TABLE departments (
    department_id NUMBER(4) PRIMARY KEY,
    department_name VARCHAR2(30) NOT NULL,
    manager_id NUMBER(6),
    location_id NUMBER(4)
);
```

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	DEPARTMENT_ID	NUMBER(4,0)	No	(null)	1	(null)
2	DEPARTMENT_NAME	VARCHAR2(30 BYTE)	No	(null)	2	(null)
3	MANAGER_ID	NUMBER(6,0)	Yes	(null)	3	(null)
4	LOCATION_ID	NUMBER(4,0)	Yes	(null)	4	(null)

b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>

```
CREATE TABLE employees (
    employee_id NUMBER(6) PRIMARY KEY,
    last_name VARCHAR2(25) NOT NULL,
    email VARCHAR2(25),
    salary NUMBER(8,2),
    commission_pct NUMBER(2,2),
    hire_date DATE NOT NULL,
    department_id NUMBER(4),
    CONSTRAINT fk_dept FOREIGN KEY (department_id) REFERENCES departments (department_id)
);
```

	COLUMN_N...	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	EMPLOYEE_ID	NUMBER(6,0)	No	(null)	1	(null)
2	LAST_NAME	VARCHAR2(25 BYTE)	No	(null)	2	(null)
3	EMAIL	VARCHAR2(25 BYTE)	Yes	(null)	3	(null)
4	SALARY	NUMBER(8,2)	Yes	(null)	4	(null)
5	COMMISSION_PCT	NUMBER(2,2)	Yes	(null)	5	(null)
6	HIRE_DATE	DATE	No	(null)	6	(null)
7	DEPARTMENT_ID	NUMBER(4,0)	Yes	(null)	7	(null)

c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).

```
CREATE TABLE job_history (
    employee_id NUMBER(6) NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    job_id VARCHAR2(10) NOT NULL,
    department_id NUMBER(4),
    CONSTRAINT pk_job_hist PRIMARY KEY (employee_id, start_date),
    CONSTRAINT fk_emp_hist FOREIGN KEY (employee_id) REFERENCES employees (employee_id)
);
```

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	EMPLOYEE_ID	NUMBER(6,0)	No	(null)	1	(null)
2	START_DATE	DATE	No	(null)	2	(null)
3	END_DATE	DATE	No	(null)	3	(null)
4	JOB_ID	VARCHAR2(10 BYTE)	No	(null)	4	(null)
5	DEPARTMENT_ID	NUMBER(4,0)	Yes	(null)	5	(null)

```
CREATE TABLE locations (
    location_id NUMBER(4) PRIMARY KEY,
    street_address VARCHAR2(40),
    city VARCHAR2(30) NOT NULL,
    state_province VARCHAR2(25)
);
```

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	LOCATION_ID	NUMBER(4,0)	No	(null)	1	(null)
2	STREET_ADDRESS	VARCHAR2(40 BYTE)	Yes	(null)	2	(null)
3	CITY	VARCHAR2(30 BYTE)	No	(null)	3	(null)
4	STATE_PROVINCE	VARCHAR2(25 BYTE)	Yes	(null)	4	(null)

```
CREATE TABLE suppliers (
    supplier_id NUMBER(6) PRIMARY KEY,
    supplier_name VARCHAR2(50) NOT NULL,
    contact_name VARCHAR2(50),
    city VARCHAR2(30)
);
```

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	SUPPLIER_ID	NUMBER(6,0)	No	(null)	1	(null)
2	SUPPLIER_NAME	VARCHAR2(50 BYTE)	No	(null)	2	(null)
3	CONTACT_NAME	VARCHAR2(50 BYTE)	Yes	(null)	3	(null)
4	CITY	VARCHAR2(30 BYTE)	Yes	(null)	4	(null)

2. Analisis table-table dengan compute statistics

```
ANALYZE TABLE departments COMPUTE STATISTICS;
ANALYZE TABLE employees COMPUTE STATISTICS;
ANALYZE TABLE suppliers COMPUTE STATISTICS;
ANALYZE TABLE locations COMPUTE STATISTICS;
ANALYZE TABLE job history COMPUTE STATISTICS;
```

3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).

   a. Locations

```
INSERT ALL
  INTO locations VALUES (1000, '1297 Via Cola di Rie', 'Roma', 'IT')
  INTO locations VALUES (1100, '93091 Calle della Testa', 'Venice', 'IT')
  INTO locations VALUES (1200, '2017 Shinjuku-ku', 'Tokyo', 'Tokyo')
  INTO locations VALUES (1300, '9450 Kamiya-cho', 'Hiroshima', 'JP')
  INTO locations VALUES (1400, '2014 Jabberwocky Rd', 'Southlake', 'Texas')
  INTO locations VALUES (1500, '2011 Interiors Blvd', 'South San Francisco', 'California')
  INTO locations VALUES (1600, '2007 Zagora St', 'South Brunswick', 'New Jersey')
  INTO locations VALUES (1700, '2004 Charade Rd', 'Seattle', 'Washington')
  INTO locations VALUES (1800, '147 Spadina Ave', 'Toronto', 'Ontario')
  INTO locations VALUES (1900, '6092 Boxwood St', 'Whitehorse', 'Yukon')
SELECT * FROM dual; [cite: 83, 84, 85, 86]
```

	LOCATION_ID	STREET_ADDRESS	CITY	STATE_PROVINCE
1	1000	1297 Via Cola di Rie	Roma	IT
2	1100	93091 Calle della Testa	Venice	IT
3	1200	2017 Shinjuku-ku	Tokyo	Tokyo
4	1300	9450 Kamiya-cho	Hiroshima	JP
5	1400	2014 Jabberwocky Rd	Southlake	Texas
6	1500	2011 Interiors Blvd	South San Francisco	California
7	1600	2007 Zagora St	South Brunswick	New Jersey
8	1700	2004 Charade Rd	Seattle	Washington
9	1800	147 Spadina Ave	Toronto	Ontario
10	1900	6092 Boxwood St	Whitehorse	Yukon

   b. Departments

```
INSERT ALL
  INTO departments VALUES (10, 'Administration', 200, 1700)
  INTO departments VALUES (20, 'Marketing', 201, 1800)
  INTO departments VALUES (30, 'Purchasing', 114, 1700)
  INTO departments VALUES (40, 'Human Resources', 203, 1400)
  INTO departments VALUES (50, 'Shipping', 121, 1500)
  INTO departments VALUES (60, 'IT', 103, 1400)
  INTO departments VALUES (70, 'Public Relations', 204, 1700)
  INTO departments VALUES (80, 'Sales', 145, 1500)
  INTO departments VALUES (90, 'Executive', 100, 1700)
  INTO departments VALUES (100, 'Finance', 108, 1700)
SELECT * FROM dual;
```

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	Administration	200	1700
2	20	Marketing	201	1800
3	60	IT	103	1400
4	30	Purchasing	114	1700
5	40	Human Resources	203	1400
6	50	Shipping	121	1500
7	70	Public Relations	204	1700
8	80	Sales	145	1500
9	90	Executive	100	1700
10	100	Finance	108	1700

c. Employees

```
INSERT ALL
  INTO employees VALUES (101, 'Kochhar', 'NHOCHHAR', 17000, NULL, TO_DATE('2021-09-21', 'YYYY-MM-DD'), 90)
  INTO employees VALUES (102, 'De Haan', 'LDEHAAN', 17000, NULL, TO_DATE('2022-01-13', 'YYYY-MM-DD'), 90)
  INTO employees VALUES (103, 'Hunold', 'AHUNOLD', 9000, NULL, TO_DATE('2022-01-03', 'YYYY-MM-DD'), 60)
  INTO employees VALUES (104, 'Ernst', 'AERNST', 6000, NULL, TO_DATE('2023-05-21', 'YYYY-MM-DD'), 60)
  INTO employees VALUES (107, 'Lorentz', 'DLORENTZ', 4200, NULL, TO_DATE('2024-02-07', 'YYYY-MM-DD'), 60)
  INTO employees VALUES (145, 'Russell', 'JRUSSELL', 14000, 0.4, TO_DATE('2021-10-01', 'YYYY-MM-DD'), 80)
  INTO employees VALUES (146, 'Partners', 'KPARTNER', 13500, 0.3, TO_DATE('2021-01-05', 'YYYY-MM-DD'), 80)
  INTO employees VALUES (147, 'Errazuriz', 'AERRAZUR', 12000, 0.3, TO_DATE('2022-03-10', 'YYYY-MM-DD'), 80)
  INTO employees VALUES (148, 'John', 'GCAMBRAU', 11000, 0.3, TO_DATE('2023-10-15', 'YYYY-MM-DD'), 80)
  INTO employees VALUES (149, 'Peter', 'ELZLOTKE', 10500, 0.2, TO_DATE('2024-01-29', 'YYYY-MM-DD'), 80)
SELECT * FROM dual;
```

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID
1	101	Kochhar	NHOCHHAR	17000	(null)	21-SEP-21	90
2	102	De Haan	LDEHAAN	17000	(null)	13-JAN-22	90
3	103	Hunold	AHUNOLD	9000	(null)	03-JAN-22	60
4	104	Ernst	AERNST	6000	(null)	21-MAY-23	60
5	107	Lorentz	DLORENTZ	4200	(null)	07-FEB-24	60
6	145	Russell	JRUSSELL	14000	0.4	01-OCT-21	80
7	146	Partners	KPARTNER	13500	0.3	05-JAN-21	80
8	147	Errazuriz	AERRAZUR	12000	0.3	10-MAR-22	80
9	148	John	GCAMBRAU	11000	0.3	15-OCT-23	80
10	149	Peter	ELZLOTKE	10500	0.2	29-JAN-24	80

d. Job_History

```
INSERT ALL
  INTO job_history VALUES (101, TO_DATE('2015-09-21', 'YYYY-MM-DD'), TO_DATE('2016-10-27', 'YYYY-MM-DD'), 'AC_ACCOUNT', 110)
  INTO job_history VALUES (101, TO_DATE('2016-10-28', 'YYYY-MM-DD'), TO_DATE('2019-03-15', 'YYYY-MM-DD'), 'AC_MGR', 110)
  INTO job_history VALUES (102, TO_DATE('2011-01-13', 'YYYY-MM-DD'), TO_DATE('2016-07-24', 'YYYY-MM-DD'), 'IT_PROG', 60)
  INTO job_history VALUES (103, TO_DATE('2020-01-01', 'YYYY-MM-DD'), TO_DATE('2021-12-31', 'YYYY-MM-DD'), 'IT_PROG', 60)
  INTO job_history VALUES (104, TO_DATE('2021-01-01', 'YYYY-MM-DD'), TO_DATE('2022-12-31', 'YYYY-MM-DD'), 'ST_CLERK', 50)
  INTO job_history VALUES (107, TO_DATE('2022-01-01', 'YYYY-MM-DD'), TO_DATE('2023-12-31', 'YYYY-MM-DD'), 'SA_REP', 80)
  INTO job_history VALUES (145, TO_DATE('2018-01-01', 'YYYY-MM-DD'), TO_DATE('2019-12-31', 'YYYY-MM-DD'), 'SA_MAN', 80)
  INTO job_history VALUES (146, TO_DATE('2019-01-01', 'YYYY-MM-DD'), TO_DATE('2020-12-31', 'YYYY-MM-DD'), 'SA_REP', 80)
  INTO job_history VALUES (147, TO_DATE('2020-01-01', 'YYYY-MM-DD'), TO_DATE('2021-12-31', 'YYYY-MM-DD'), 'SA_REP', 80)
  INTO job_history VALUES (148, TO_DATE('2021-01-01', 'YYYY-MM-DD'), TO_DATE('2022-12-31', 'YYYY-MM-DD'), 'SA_REP', 80)
SELECT * FROM dual;
```

	EMPLOYEE_ID	START_DATE	END_DATE	JOB_ID	DEPARTMENT_ID
1	101	21-SEP-15	27-OCT-16	AC_ACCOUNT	110
2	101	28-OCT-16	15-MAR-19	AC_MGR	110
3	102	13-JAN-11	24-JUL-16	IT_PROG	60
4	103	01-JAN-20	31-DEC-21	IT_PROG	60
5	104	01-JAN-21	31-DEC-22	ST_CLERK	50
6	107	01-JAN-22	31-DEC-23	SA_REP	80
7	145	01-JAN-18	31-DEC-19	SA_MAN	80
8	146	01-JAN-19	31-DEC-20	SA_REP	80
9	147	01-JAN-20	31-DEC-21	SA_REP	80
10	148	01-JAN-21	31-DEC-22	SA_REP	80

e. Suplliers

Worksheet | Query Builder

```
INSERT ALL
  INTO suppliers VALUES (1, 'Logitech', 'Mark', 'Lausanne')
  INTO suppliers VALUES (2, 'Microsoft', 'Bill', 'Redmond')
  INTO suppliers VALUES (3, 'Oracle Corp', 'Larry', 'Redwood City')
  INTO suppliers VALUES (4, 'Intel', 'Pat', 'Santa Clara')
  INTO suppliers VALUES (5, 'Samsung', 'Lee', 'Seoul')
  INTO suppliers VALUES (6, 'Sony', 'Ken', 'Tokyo')
  INTO suppliers VALUES (7, 'Dell', 'Michael', 'Round Rock')
  INTO suppliers VALUES (8, 'HP', 'Enrique', 'Palo Alto')
  INTO suppliers VALUES (9, 'Cisco', 'Chuck', 'San Jose')
  INTO suppliers VALUES (10, 'Lenovo', 'Yang', 'Beijing')
SELECT * FROM dual;
```

	SUPPLI...	SUPPLIER_NAME	CONTACT_NAME	CITY
1	1	Logitech	Mark	Lausanne
2	2	Microsoft	Bill	Redmond
3	3	Oracle Corp	Larry	Redwood City
4	4	Intel	Pat	Santa Clara
5	5	Samsung	Lee	Seoul
6	6	Sony	Ken	Tokyo
7	7	Dell	Michael	Round Rock
8	8	HP	Enrique	Palo Alto
9	9	Cisco	Chuck	San Jose
10	10	Lenovo	Yang	Beijing

4. Analisis table-table dengan compute statistics

```
ANALYZE TABLE departments COMPUTE STATISTICS;
ANALYZE TABLE employees COMPUTE STATISTICS;
ANALYZE TABLE suppliers COMPUTE STATISTICS;
ANALYZE TABLE locations COMPUTE STATISTICS;
ANALYZE TABLE job_history COMPUTE STATISTICS;
```

5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.

```
CREATE OR REPLACE VIEW empvu80 AS
SELECT employee_id AS id_number, last_name AS name, salary, department_id
FROM employees
WHERE department_id = 80;
```

6. Kemudian tampilkan struktur dari view empvu80

Query Result

SQL | All Rows Fetched: 5 in 0.006 seconds

	ID_NUMBER	NAME	SALARY	DEPARTMENT_ID
1	145	Russell	14000	80
2	146	Partners	13500	80
3	147	Errazuriz	12000	80
4	148	John	11000	80
5	149	Peter	10500	80

7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE

```
CREATE INDEX emp_dept_idx ON employees (department_id);
```

```
Name           Null?    Type
-------------- -------- ------------
ID_NUMBER      NOT NULL NUMBER(6)
NAME           NOT NULL VARCHAR2(25)
SALARY                  NUMBER(8,2)
DEPARTMENT_ID           NUMBER(4)
```